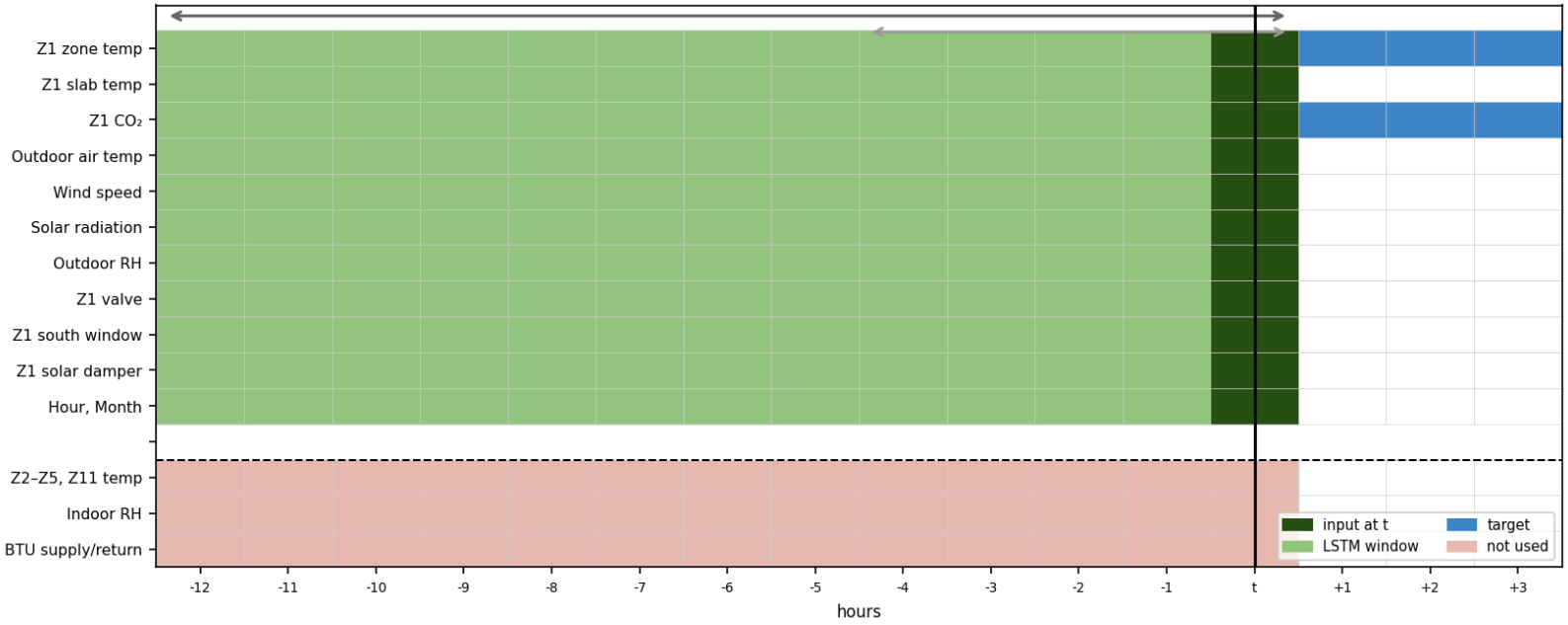
Chen et al. (2023): EMA-MPC

- Two stacked-LSTM at different time scales:
  - f_{fast}: 1 min steps, 15-step lookback
  - f_{slow}: 15 min steps, 10-step lookback (4h45min)
- 5 h prediction horizon, ensembled
- Bound control: $T_{\text{room}} > 19°\text{C}$
- MAE reduction vs single LSTM: **78.6 %** (CO_2), **88.9 %** (T_{room})

Setup

	Chen et al.	**This study**
Resolution	1 min / 15 min	1 h
Source	Modelica simulator	HouseZero sensors
Duration	3 months	1 year (Jun 2023–May 2024)
Horizon	5 h (recursive)	1 h, 2 h, 3 h (direct)
$\text{LSTM}_{\text{fast}}$	10 steps (10 min)	4 steps (4 h)
$\text{LSTM}_{\text{slow}}$	10 steps (150 min)	12 steps (12 h)
Evaluation	70/20/10 split	Leave-one-month-out (12 folds)
Features	10–11	14
Targets	T_{room}, CO_2	T_{room}, CO_2

Model inputs and targets



$\text{LSTM}_{\text{slow}}$: 12 h lookback $\text{LSTM}_{\text{fast}}$: 4 h lookback Ridge/XGBoost: features at t only (+ explicit lags $t-1$, $t-24$)

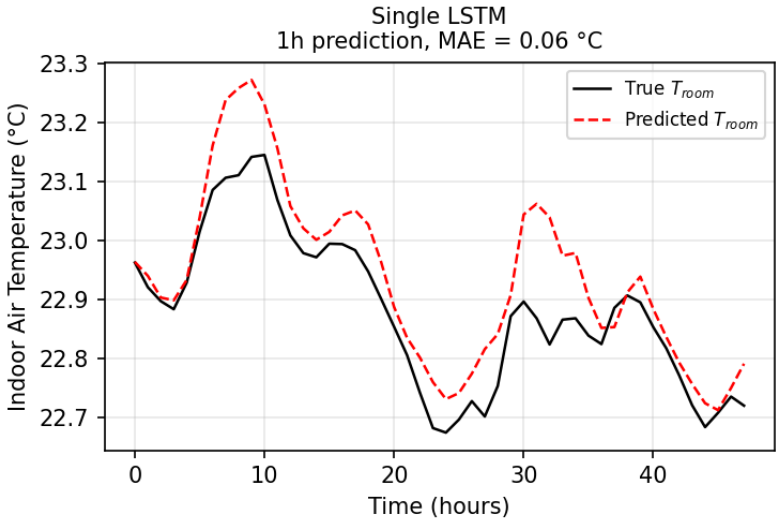
MAE comparison

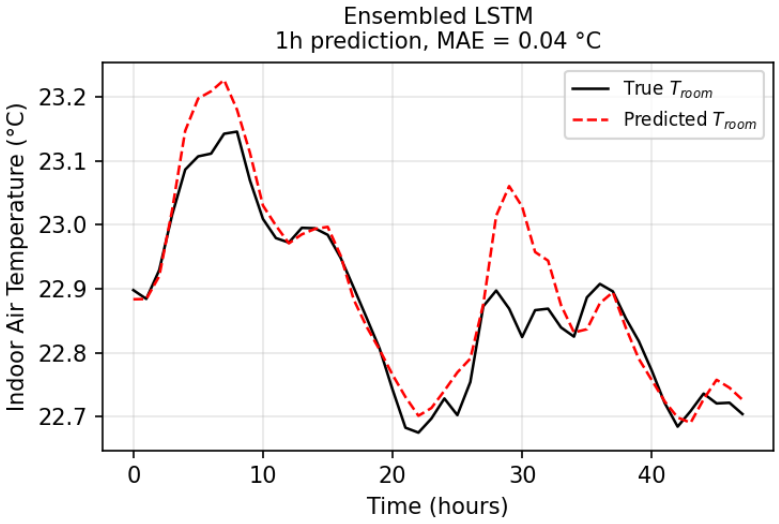
T_{room} MAE (°C)	1 h	2 h	3 h
Persistence	0.093	0.166	0.229
Ridge	**0.076**	0.147	0.208
XGBoost	0.080	**0.141**	**0.194**
LSTM	0.115	0.189	0.243
Ens-LSTM	0.104	0.187	0.247

CO_2 MAE (ppm)	1 h	2 h	3 h
Persistence	22.6	40.8	55.8
Ridge	**24.1**	42.4	55.6
XGBoost	24.8	41.0	**52.6**
LSTM	27.2	44.1	55.3
Ens-LSTM	25.1	**42.1**	54.0

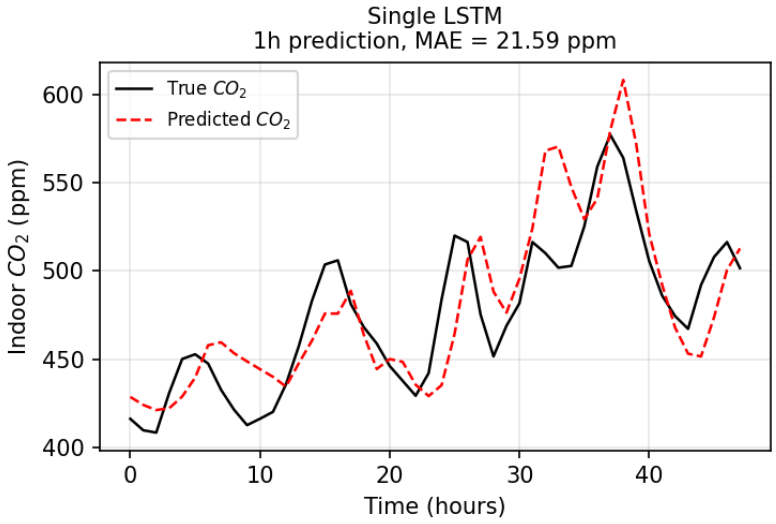
	T_{room} RMSE / R^2			CO_2 RMSE / R^2		
	1 h	2 h	3 h	1 h	2 h	3 h
XGBoost	**.131** / .940	**.220** / .832	**.290** / .711	55.6 / .822	79.2 / .619	95.0 / .420
Ens-LSTM	.162 / .903	.271 / .734	.346 / .563	**52.3** / .844	**77.0** / .651	**91.8** / .491

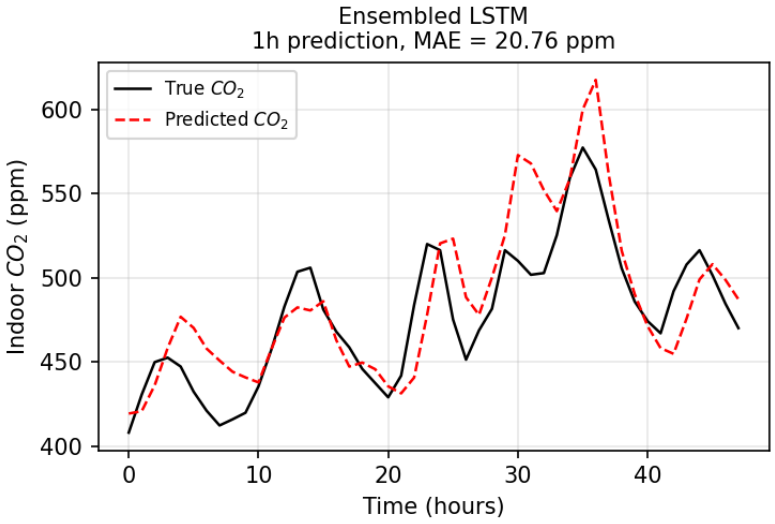
Single LSTM vs Ensembled LSTM: T_{room} (1 h)



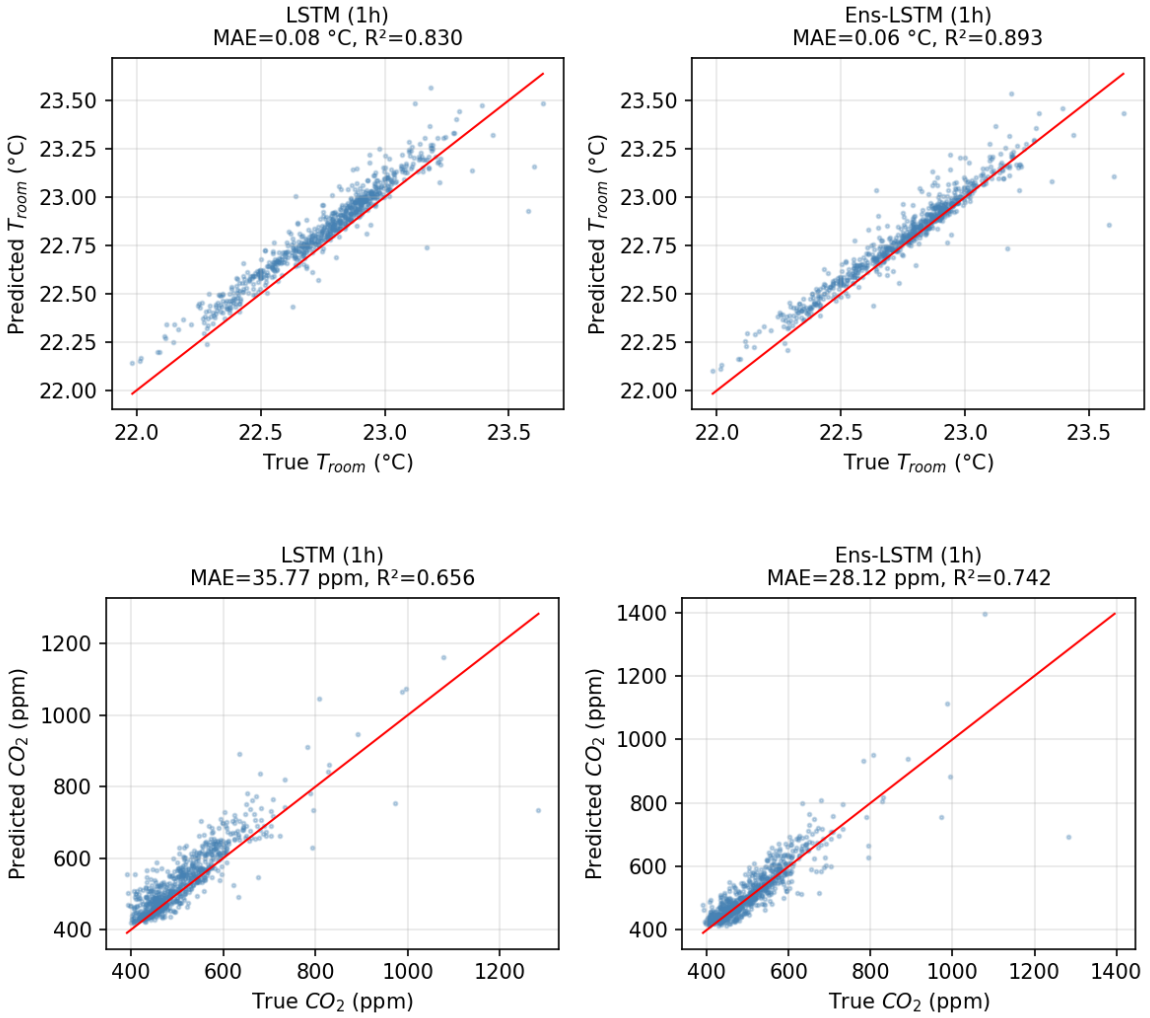


Single LSTM vs Ensembled LSTM: CO_2 (1 h)

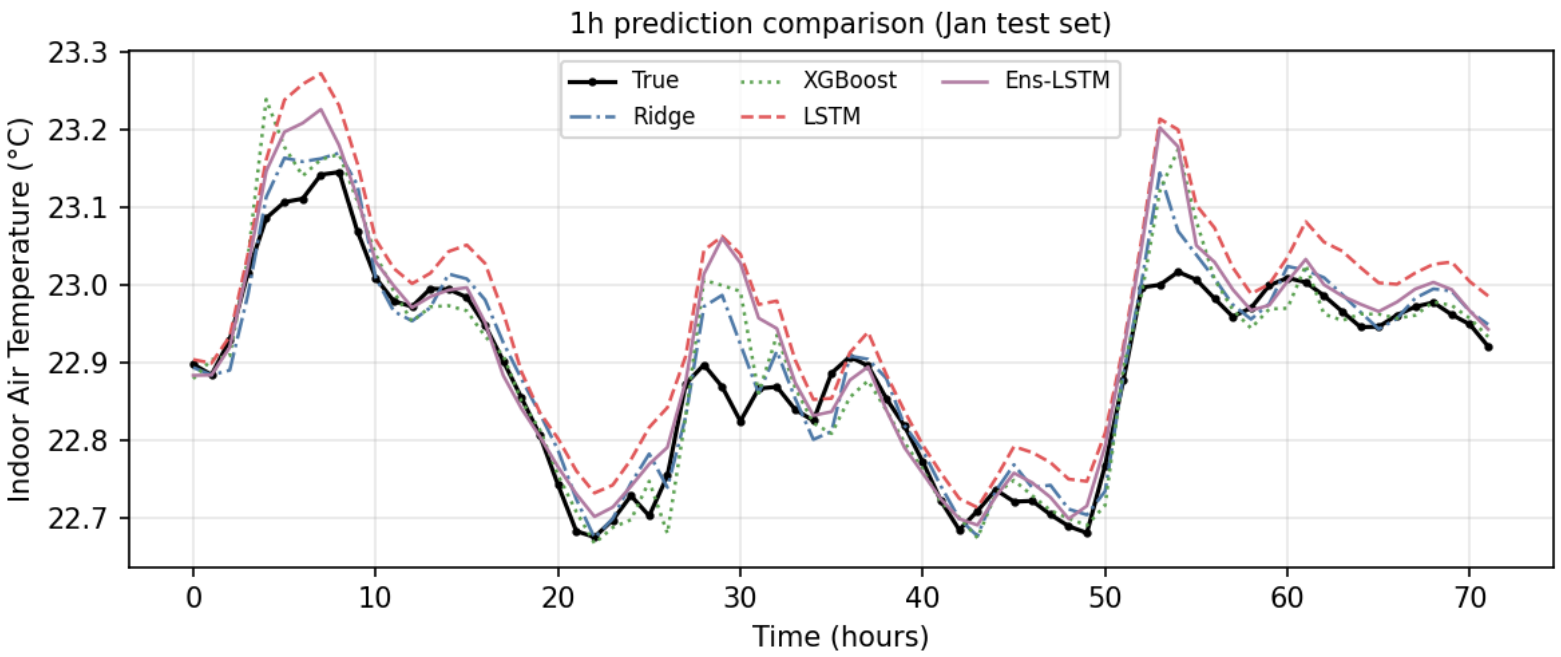




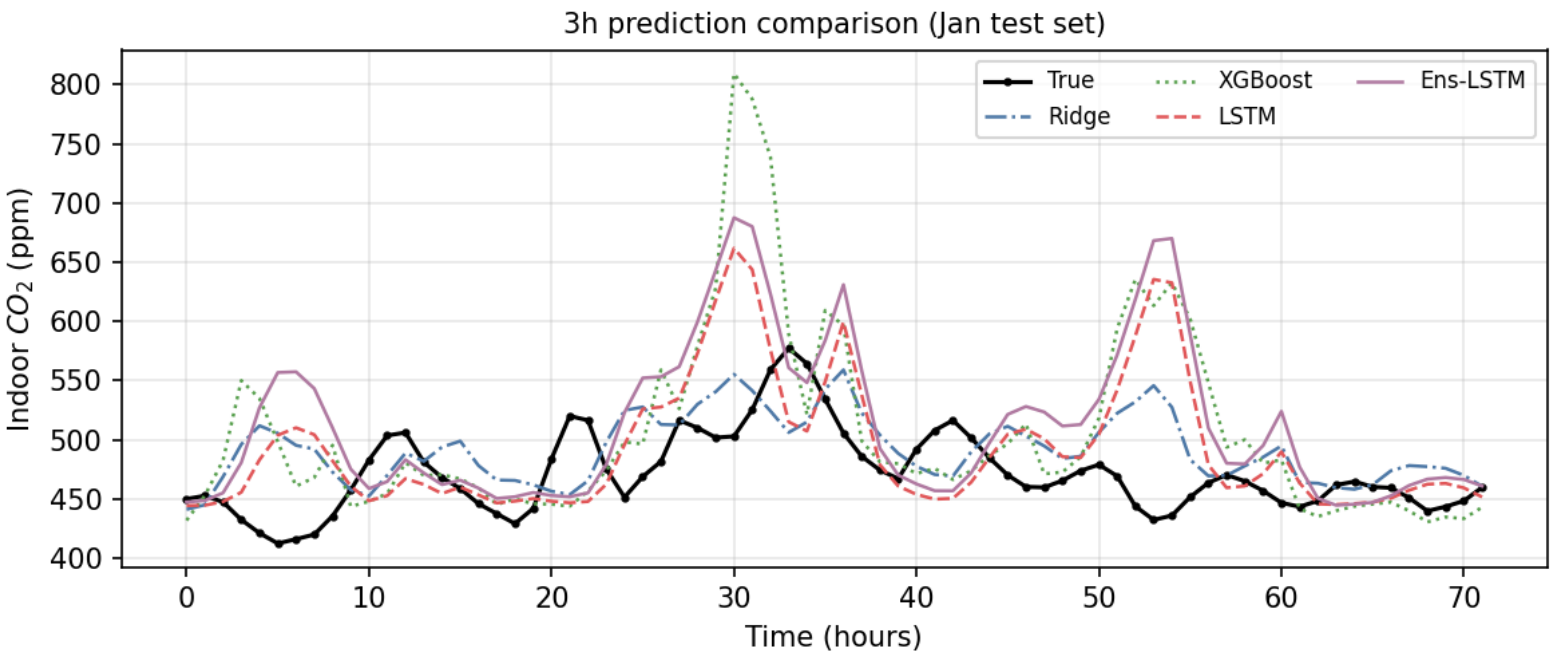
Predicted vs true (1 h)



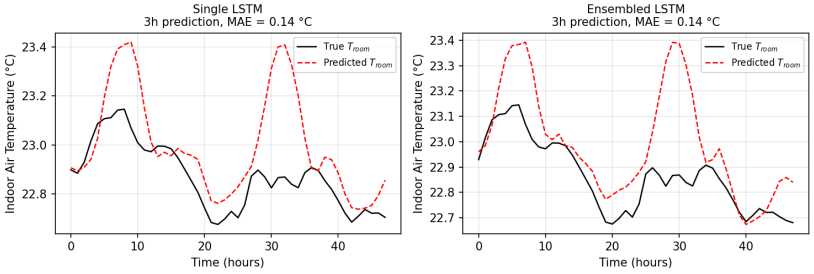
Model comparison: T_{room} (1 h)

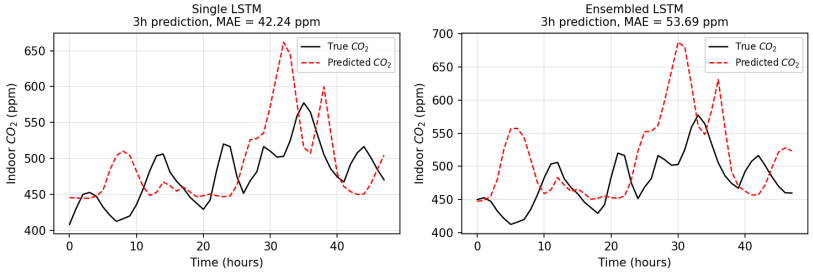


Model comparison: CO_2 (3 h)

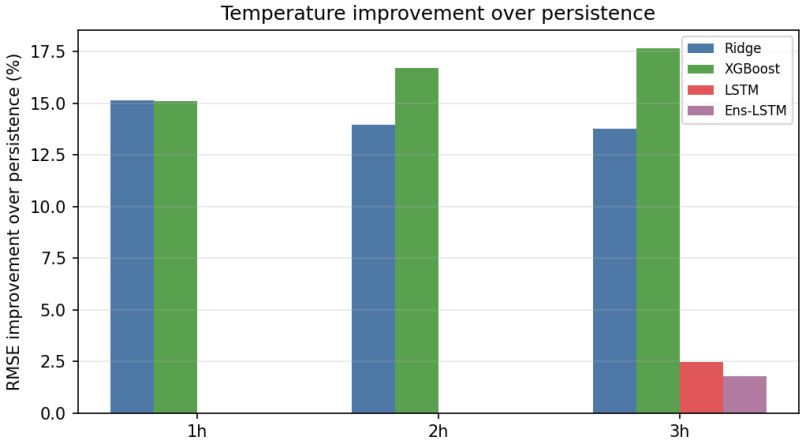


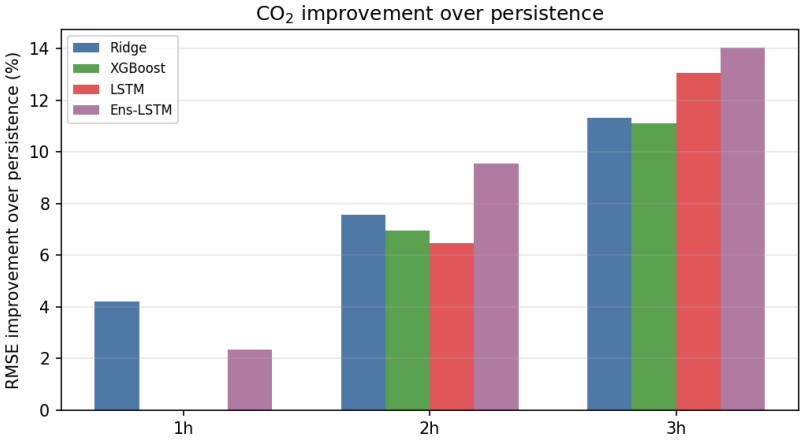
3 h horizon


Single LSTM
3h prediction, MAE = 0.14 °C
Ensembled LSTM
3h prediction, MAE = 0.14 °C
Indoor Air Temperature (°C)
Time (hours)
True T_{room}
Predicted T_{room}


Single LSTM
3h prediction, MAE = 42.24 ppm
Ensembled LSTM
3h prediction, MAE = 53.69 ppm
Indoor CO_2 (ppm)
Time (hours)
True CO_2
Predicted CO_2

Improvement over persistence





Summary

$T_{\mathbf{room}}$:

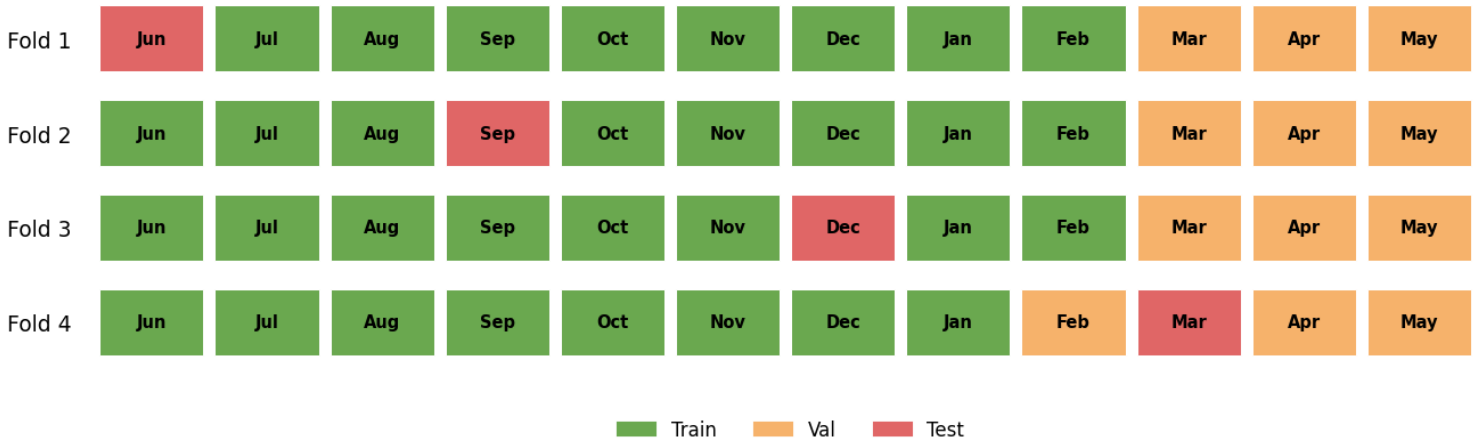
- XGBoost best: 0.131°C RMSE, $R^2 = 0.94$ (1 h)
- LSTM: 0.162°C — hourly resolution limits multi-scale advantage

$\mathbf{CO_2}$:

- Ens-LSTM best: 77.0 ppm RMSE (2 h), 91.8 ppm (3 h)
- Captures fast spikes + slow trends

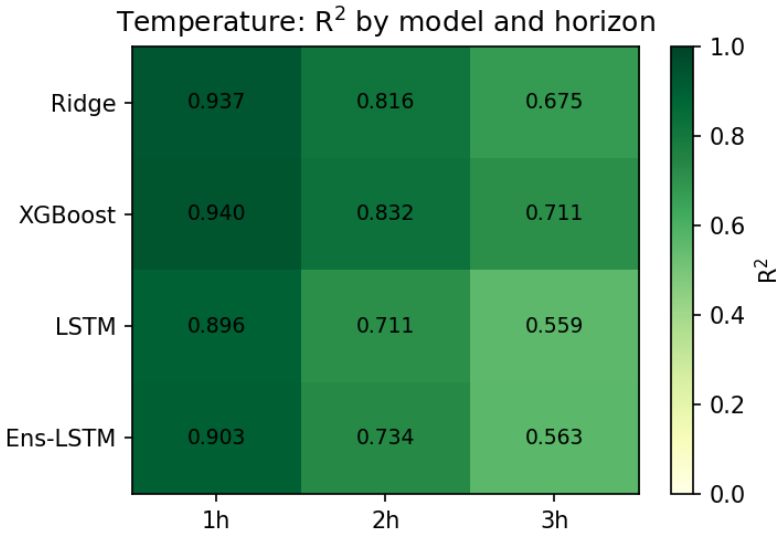
Backup

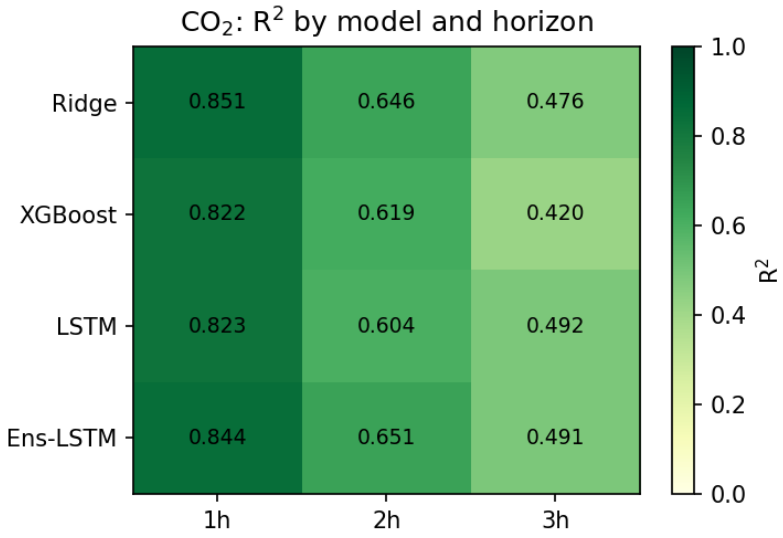
Cross-validation split



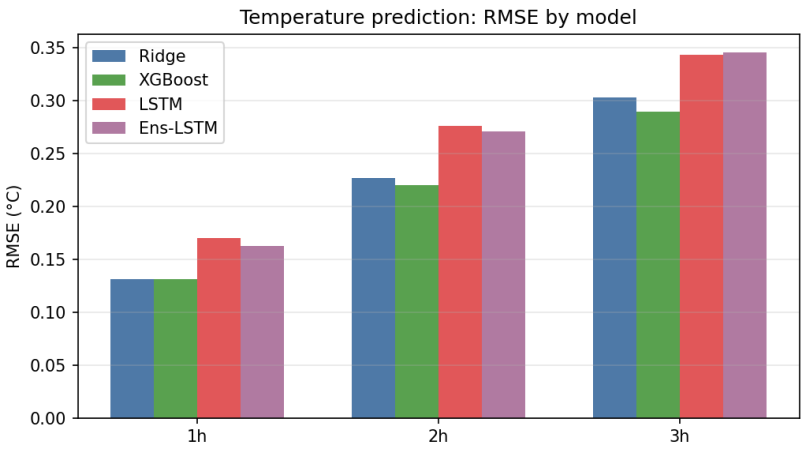
12 folds $\times$ 12 months. Per fold: $\sim$7990 train, $\sim$730 test. Last 20% of train $\rightarrow$ validation (LSTM early stopping). 4 of 12 folds shown.

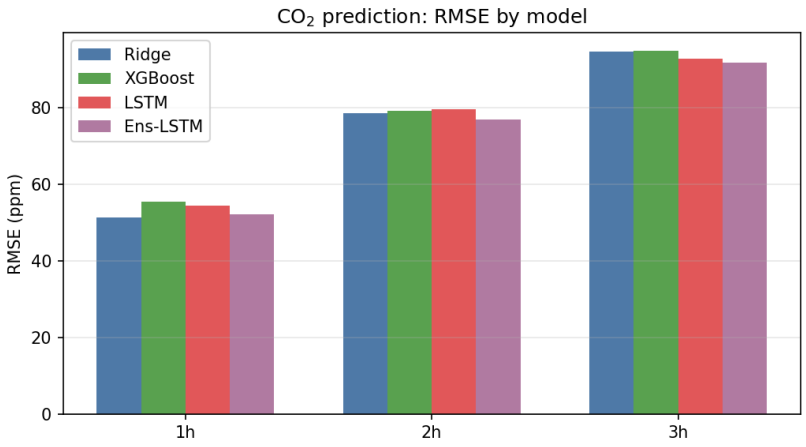
R^2 heatmaps


Temperature: R2 by model and horizon
Ridge
XGBoost
LSTM
Ens-LSTM
0.937
0.816
0.675
0.940
0.832
0.711
0.896
0.711
0.559
0.903
0.734
0.563
1h
2h
3h
1.0
0.8
0.6
0.4
0.2
0.0
R2


CO2: R2 by model and horizon
Ridge
XGBoost
LSTM
Ens-LSTM
0.851
0.646
0.476
0.822
0.619
0.420
0.823
0.604
0.492
0.844
0.651
0.491
1h
2h
3h
1.0
0.8
0.6
0.4
0.2
0.0
R2

RMSE by model





Abbreviations and reference

LSTM	Long Short-Term Memory (recurrent neural network)
Ens-LSTM	Ensembled LSTM (fast + slow timescale)
EMA-MPC	Ensembled Multi-time scale Adaptive Model Predictive Control
MPC	Model Predictive Control
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R^2	Coefficient of determination
CV	Cross-Validation
BTU	British Thermal Unit (energy meter)
RH	Relative Humidity

Reference:

E.X. Chen, X. Han, A. Malkawi, R. Zhang, N. Li. "Adaptive model predictive control with ensembled multi-time scale deep-learning models for smart control of natural ventilation." *Building and Environment*, 242:110519, 2023.
`doi:10.1016/j.buildenv.2023.110519`